THE POWER OF VACCINATION

From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of how vaccines transformed public health by eliminating or drastically reducing deadly diseases over the past two centuries.

1796–2020
scope of vaccine history highlighted

Evidence → action
audience: public health & policy

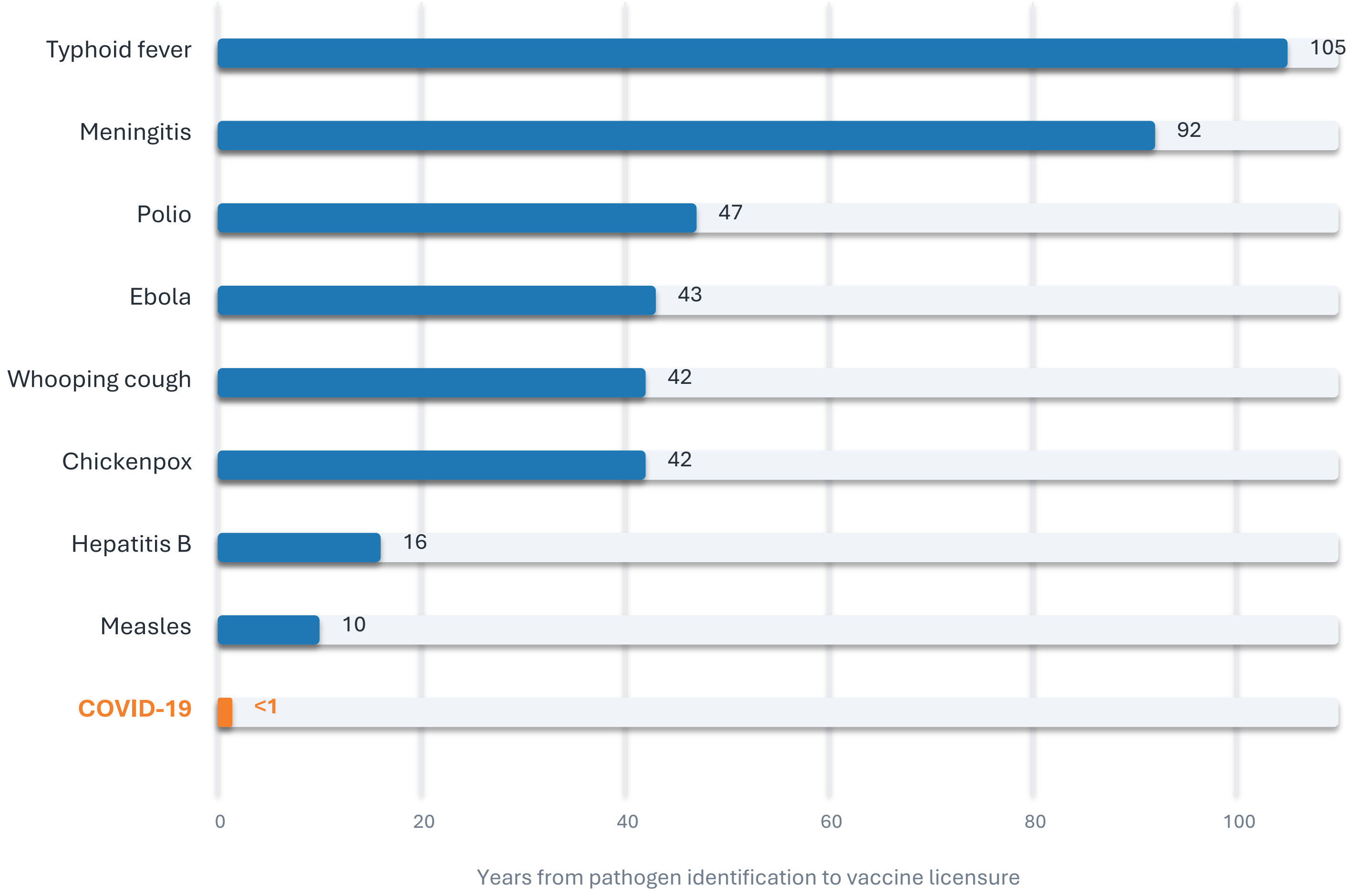
1796
Smallpox first vaccine

1980
Smallpox eradicated

2020
COVID-19 licensed

Source: Our World in Data vaccination brief

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



105 years
early benchmark from brief

<1 year
COVID-19 timeline

- The brief links faster timelines to advanced microscopy, genome sequencing, mRNA technology, and virus-like particles.
- Acceleration does not erase the policy challenge: access, coverage, and trust still determine impact.

Timeline data: pathogen identification → vaccine licensure, as reported in source brief.

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-vaccine cases	Post-vaccine cases	Case reduction	Pre-vaccine deaths	Death reduction
Diphtheria	158	0	**100%**	13.7	**100%**
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	**100%**	2.9	**100%**
Acute polio	141	0	**100%**	10	**100%**
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	**40.5%**	24	**31.3%**

✓ 100% case and death reduction: diphtheria, smallpox, acute polio

Remaining burden: pneumococcal disease, hepatitis A, acute hepatitis B

Counts are per million/year from Roush and Murphy (2007), as summarized in the brief.

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

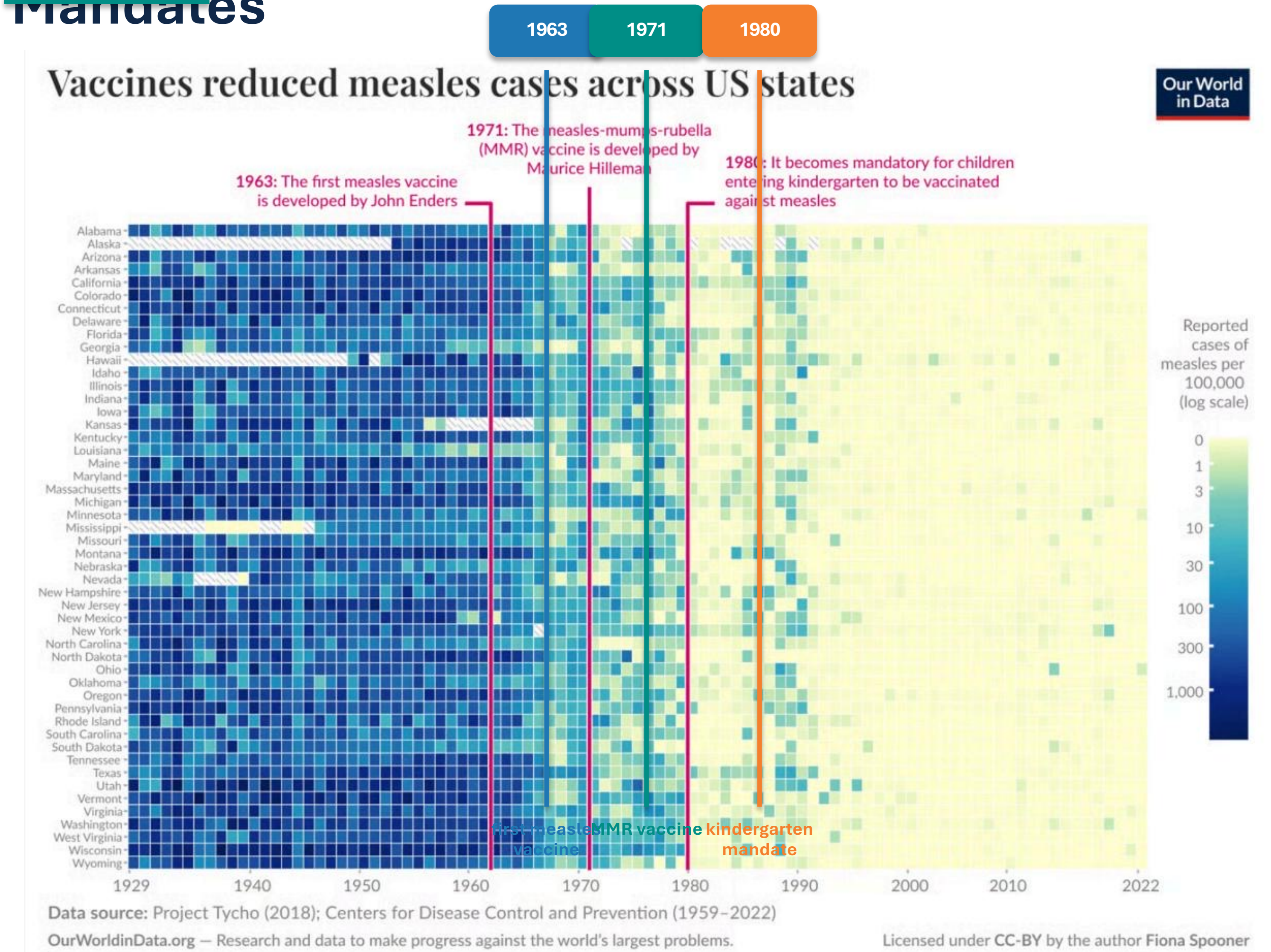


Figure reference: measles cases per 100,000 across all 50 US states, 1929–2022.

Download image or data

What the heatmap shows

- Before 1963, nearly every state saw high measles incidence: 100–1,000+ cases per 100,000.
- After the 1980 kindergarten mandate, cases dropped to near zero across all states.
- The post-mandate pattern is not zero risk: the brief notes only sporadic outbreaks.

near zero
reported post-mandate state pattern

Smallpox: The Only Human Disease Eradicated Through Vaccination

1959	1967	1977	1980
WHO resolution to eradicate	intensified campaign	last natural case in Somalia	eradication certified

eradicated worldwide
global status from brief

only human disease
uniqueness

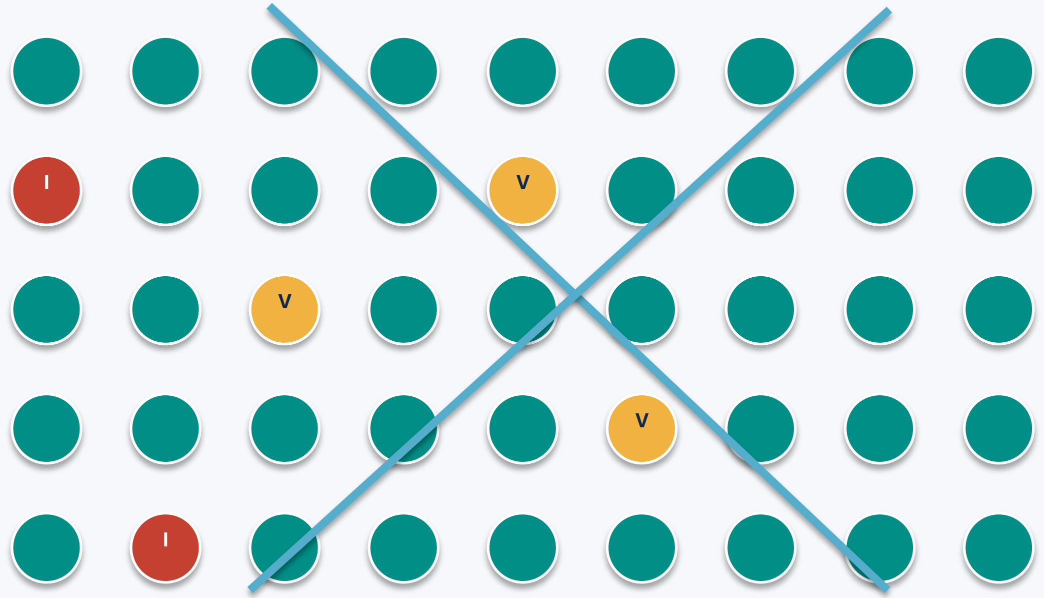
Why eradication was possible

- The virus infected only humans.
- Symptoms were distinguishable.
- The vaccine was cheap and effective.

Narrative source: WHO resolution in 1959, intensified campaign in 1967, last natural case in Somalia in 1977, certification in 1980.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Population-level shielding



Vaccinated people interrupt transmission pathways

Vaccinated

Vulnerable

Infected

Mechanism

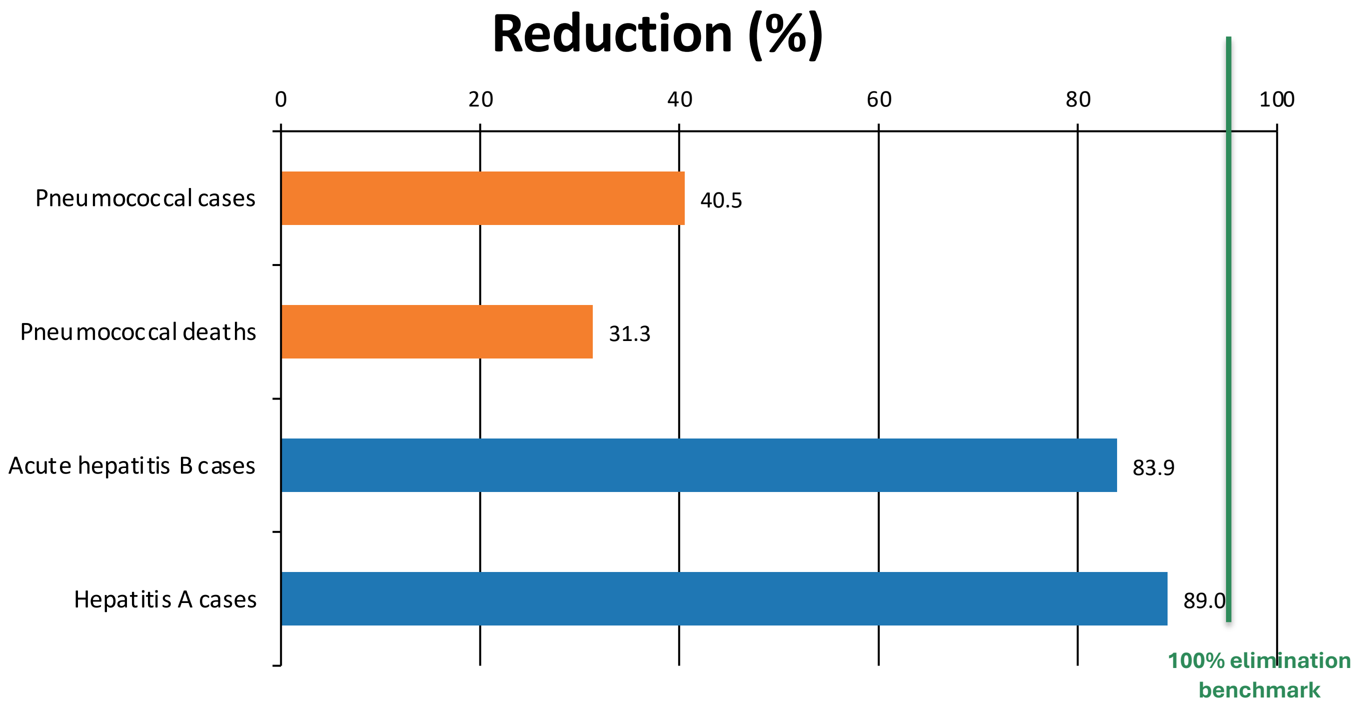
When enough people are vaccinated, pathogens cannot spread effectively.

That community-level protection shields people who cannot be vaccinated:

- newborns
- immunocompromised individuals
- cancer patients on chemotherapy

Implication for policy: vaccination protects individuals and the surrounding network.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



Barriers named in the brief

Access gaps

Not everyone has access to vaccines globally.

Supply shortages

Shortages put children at risk.

Trust & understanding

Poor scientific understanding can reduce uptake.

40.5%
lowest case reduction highlighted

The brief also reports hepatitis A death reduction of 88.7% and acute hepatitis B death reduction of 83.6%.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

Elimination is real
100% elimination of cases and deaths for diphtheria, smallpox, and both forms of polio in the US.

Timelines collapsed
Development timelines moved from over 100 years for typhoid to under 1 year for COVID-19.

Mandates matter
Measles dropped to near zero across all 50 states after the 1980 kindergarten mandate.

Herd immunity scales
Vaccination protects communities, including newborns, immunocompromised individuals, and patients on chemotherapy.

Gaps persist
Pneumococcal disease, hepatitis A, and hepatitis B show incomplete control.

Call to action for global health systems

Improve access, protect supply, expand coverage, and build trust in vaccination.

All statistics and dates are drawn from the provided Our World in Data vaccination brief.